

Questions

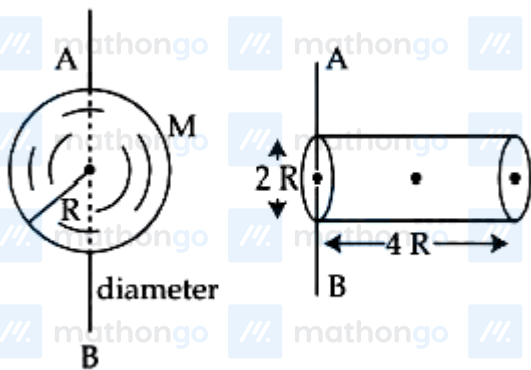
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Q1 - 2024 (04 Apr Shift 1)

A solid sphere and a hollow cylinder roll up without slipping on same inclined plane with same initial speed v . The sphere and the cylinder reaches upto maximum heights h_1 and h_2 , respectively, above the initial level. The ratio $h_1 : h_2$ is $\frac{n}{10}$. The value of n is _____.

Q2 - 2024 (05 Apr Shift 1)

Ratio of radius of gyration of a hollow sphere to that of a solid cylinder of equal mass, for moment of Inertia about their diameter axis AB as shown in figure is $\sqrt{8/x}$. The value of x is :



(1) 51

(2) 34

(3) 17

(4) 67

Q3 - 2024 (05 Apr Shift 2)

A hollow sphere is rolling on a plane surface about its axis of symmetry. The ratio of rotational kinetic energy to its total kinetic energy is $\frac{x}{5}$. The value of x is _____.

Q4 - 2024 (06 Apr Shift 1)

If the radius of earth is reduced to three-fourth of its present value without change in its mass then value of duration of the day of earth will be _____ hours 30 minutes.

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Q5 - 2024 (06 Apr Shift 2)

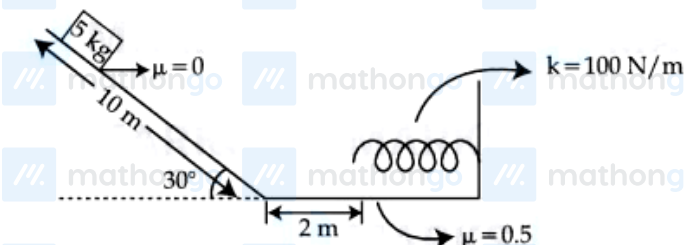
Three balls of masses 2 kg, 4 kg and 6 kg respectively are arranged at centre of the edges of an equilateral triangle of side 2 m. The moment of inertia of the system about an axis through the centroid and perpendicular to the plane of triangle, will be _____ kgm^2 .

Q6 - 2024 (08 Apr Shift 2)

A thin circular disc of mass M and radius R is rotating in a horizontal plane about an axis passing through its centre and perpendicular to its plane with angular velocity ω . If another disc of same dimensions but of mass $M/2$ is placed gently on the first disc co-axially, then the new angular velocity of the system is :

- (1) $\frac{3}{2}\omega$
- (2) $\frac{5}{4}\omega$
- (3) $\frac{2}{3}\omega$
- (4) $\frac{4}{5}\omega$

Q7 - 2024 (08 Apr Shift 2)



A block is simply released from the top of an inclined plane as shown in the figure above. The maximum compression in the spring when the block hits the spring is :

- (1) $\sqrt{6} \text{ m}$
- (2) $\sqrt{5} \text{ m}$
- (3) 1 m
- (4) 2 m

Q8 - 2024 (09 Apr Shift 1)

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A heavy iron bar, of weight W is having its one end on the ground and the other on the shoulder of a person.

The bar makes an angle θ with the horizontal. The weight experienced by the person is :

(1) $W \cos \theta$

(2) $\frac{W}{2}$

(3) W

(4) $W \sin \theta$

Q9 - 2024 (09 Apr Shift 1)

A string is wrapped around the rim of a wheel of moment of inertia 0.40 kgm^2 and radius 10 cm. The wheel is free to rotate about its axis. Initially the wheel is at rest. The string is now pulled by a force of 40 N. The angular velocity of the wheel after 10 s is $x \text{ rad/s}$, where x is _____

Q10 - 2024 (09 Apr Shift 2)

A circular disc reaches from top to bottom of an inclined plane of length l . When it slips down the plane, it takes t s. When it rolls down the plane then it takes $\left(\frac{\alpha}{2}\right)^{1/2} t$ s, where α is _____

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Answer Key

Solutions

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Q1

Gain in P.E. = Loss in K.E.

$$mgh = \frac{1}{2}mv^2 \left(1 + \frac{K^2}{R^2}\right)$$

$$h \propto 1 + \frac{K^2}{R^2}$$

$$\frac{h_1}{h_2} = \frac{1 + \frac{K^2}{R^2}}{1 + 1} = \frac{7}{5 \times 2} = \frac{7}{10}$$

$$n = 7$$

Q2

$$I_{\text{sphere}} = \frac{2}{3}MR^2 = Mk_1^2$$

$$I_{\text{cylinder}} = \frac{1}{12}M(4R^2) + \frac{1}{4}MR^2 + M(2R)^2$$

$$= \frac{67}{12}MR^2 = Mk_2^2$$

$$\frac{k_1}{k_2} = \sqrt{\frac{2}{3} \cdot \frac{12}{67}} = \sqrt{\frac{8}{67}}$$

Q3

$$\frac{\frac{1}{2}I\omega^2}{\frac{1}{2}I\omega^2 + \frac{1}{2}mv^2} = \frac{\left(\frac{1}{2}\right)\left(\frac{2}{3}mR^2\right)\omega^2}{\left(\frac{1}{2}\right)\left(\frac{2}{3}mR^2\right)\omega^2 + \frac{1}{2}m(R\omega)^2}$$

$$= \frac{\frac{2}{3}}{\frac{2}{3} + 1} = \frac{2}{5}$$

$$x = 2$$

Q4

By conservation of angular momentum

$$I_1\omega_1 = I_2\omega_2$$

$$\left(\frac{2}{5}MR^2\right)\frac{2\pi}{T_1} = \frac{2}{5}M\left(\frac{3}{4}R\right)^2\frac{2\pi}{T_2}$$

$$\frac{1}{T_1} = \frac{9}{16}\frac{1}{T_2}$$

$$\frac{1}{T_2} = \frac{9}{16} \times T_1 = \frac{9}{16} \times 24\text{hr} = \frac{27}{2}\text{hr} = 13\text{hr}30\text{mins.}$$

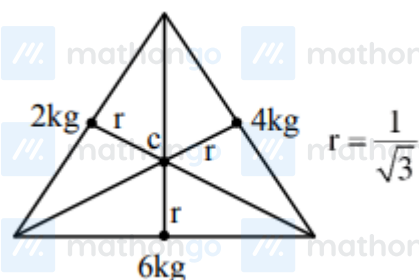
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Solutions

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Q5



Moment of inertia about C and perpendicular to the plane is :

$$I = r^2[2 + 4 + 6]$$

$$= \frac{1}{3} \times 12$$

$$I = 4 \text{ kg} - \text{m}^2$$

Q6

$$I_1 \omega = I_2 \omega_2$$

$$\frac{MR^2}{2} \omega = \frac{3}{2} \left(\frac{MR^2}{2} \right) \omega_2$$

$$\omega_2 = \frac{2}{3} \omega$$

Q7

$$w_g + w_{Fr} + w_s = \Delta KE$$

$$5 \times 10 \times 5 - 0.5 \times 5 \times 10 \times x - \frac{1}{2} Kx^2 = 0 - 0$$

$$250 = 25x + 50x^2$$

$$2x^2 + x - 10 = 0$$

$$x = 2$$

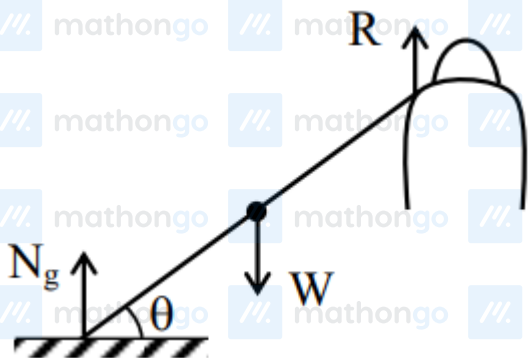
Q8

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Solutions

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R = net reaction force by shoulder

Balancing torque about pt of contact on ground:

$$W \left(\frac{L}{2} \cos \theta \right) = R(L \cos \theta)$$

$$\Rightarrow R = \frac{W}{2}$$

Q9

$$\tau = FR = I\alpha \Rightarrow 40 \times 0.1 = 0.4\alpha$$

$$\alpha = 10 \text{ rad/s}^2$$

$$W_f = 10 \times 10 = 100 \text{ rad/s}$$

Q10

For slipping

$$a = g \sin \theta$$

$$\ell = \frac{1}{2}at^2 \Rightarrow t = \sqrt{\frac{2\ell}{g \sin \theta}}$$

For rolling

$$a' = \frac{g \sin \theta}{1 + \frac{k^2}{R^2}} \left[k = \frac{R}{\sqrt{2}} \right]$$

$$\Rightarrow a' = \frac{2g \sin \theta}{3}$$

$$\ell = \frac{1}{2}a'(t')^2$$

$$\Rightarrow t' = \sqrt{\frac{6\ell}{2g \sin \theta}} = \sqrt{\frac{\alpha}{2}} \sqrt{\frac{2\ell}{g \sin \theta}}$$

$$\Rightarrow \alpha = 3$$

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